CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter delineates the research methodology employed for the design and implementation of a scalable bimodal e-voting system, with specific focus on supporting all nine Federal Constituencies of Ondo State in concurrent House of Representatives elections. The methodology is structured into three distinct phases: requirements gathering and system design; development and implementation; and testing and evaluation. This structured approach ensures that the system is robust, secure, user-centric, and aligned with the unique electoral challenges and opportunities present in the Nigerian context. The chapter details the architectural framework, the advanced biometric authentication mechanisms, the development tools and technologies selected, and the comprehensive testing strategies utilised to validate the system's efficacy and reliability.

3.2 Research Design

The research methodology for this project is systematically organised into three key phases, directly aligning with the stated research objectives and ensuring a comprehensive approach to system development and validation. Each phase builds upon the preceding one, ensuring that the final system satisfies both functional and non-functional requirements while addressing the critical gaps identified in existing bimodal e-voting solutions. Figure 3.1 summarises the three-phase structure.

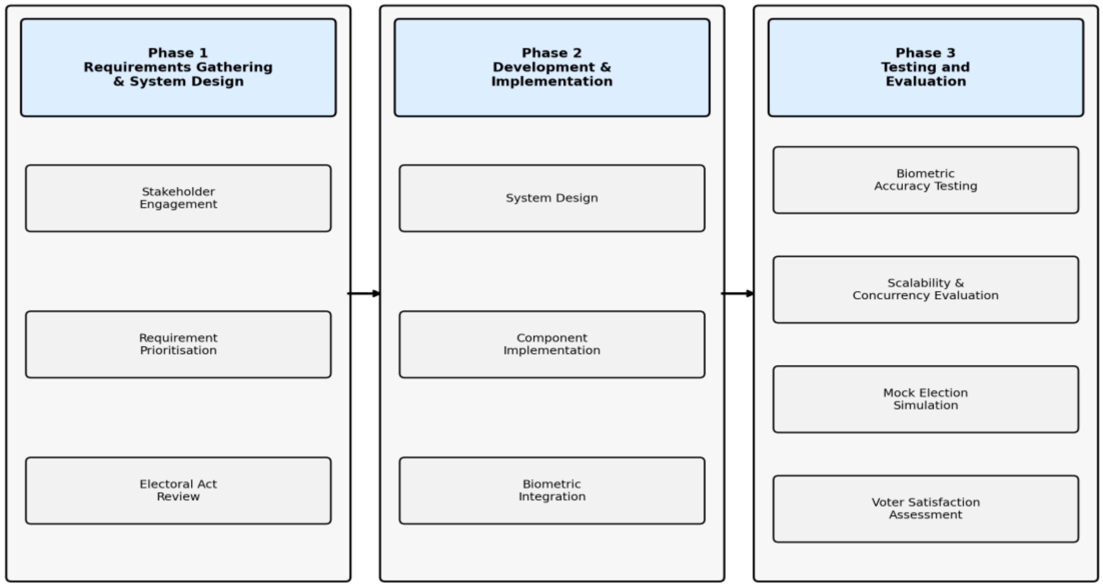

Figure 3.1: Research Methodology Phases

3.2.1 Phase 1: Requirements Gathering and System Design

This initial phase focuses on thoroughly understanding the functional and non-functional requirements essential for a successful bimodal e-voting system that can support all nine Federal Constituencies of Ondo State and their corresponding eighteen Local Government Areas simultaneously. The primary goal is to ensure the system's usability, security, scalability, and compliance with existing electoral rules and regulations. This phase involves the following activities:

a. **Stakeholder Engagement:** Identifying and consulting key stakeholders, including electoral officers, voters drawn from urban, rural, elderly, and agricultural demographics, polling officers, and collation officers. Structured questionnaires and interviews provide insights into operational requirements, anticipated challenges, and usability expectations.

b. **Requirement Prioritisation:** Identifying and prioritising critical functional requirements—including constituency-aware voter registration, independent ballot access, concurrent election management across nine constituencies, isolated result collation, offline-first biometric authentication, high-concurrency performance, and inclusive biometric fallback mechanisms (facial recognition for voters with worn fingerprints). Non-functional requirements include security, reliability, scalability, performance, user experience, accessibility, and adherence to legal frameworks.

c. **Electoral Act Review:** A meticulous review of relevant sections of the Nigerian Electoral Act (2022) and specific regulations guiding federal constituency elections in Ondo State, to ensure the system's design fully abides by legal mandates. The output of this phase is a detailed set of system specifications guiding subsequent design and implementation activities.

3.2.2 Phase 2: Development and Implementation

This phase involves the translation of gathered requirements into a concrete system architecture and the subsequent development of the scalable bimodal e-voting prototype. Primary emphasis is placed on security, the integration of advanced biometric options, offline-first capabilities, and the design of an intuitive interface for both voters and administrators.

a. **System Design:** Developing a modular layered client-server architecture that supports secure registration, constituency-aware voter authentication, vote casting, vote encryption, and result collation, with a multi-tier partitioned database structure for complete data isolation across all nine constituencies and eighteen LGAs.

b. **Component Implementation:** Building the various system modules using React Native (mobile client), Node.js with Express (backend server), and PostgreSQL with table partitioning (data layer), ensuring secure data handling and real-time interactions.

c. **Biometric Integration:** Implementing the exclusive bimodal biometric authentication mechanism via the BiometricPrompt API for fingerprint (with templates stored in the device TEE) and an on-device TensorFlow Lite pipeline comprising MobileNet-SSD detection, 68-point CNN landmark extraction, and quantised FaceNet embedding for facial recognition.

3.2.3 Phase 3: Testing and Evaluation

The final phase involves rigorous testing and evaluation of the implemented system against predefined metrics, validating its reliability, accuracy, scalability, and overall effectiveness.

a. **Biometric Accuracy Testing:** Assessing False Acceptance Rate (FAR), False Rejection Rate (FRR), Equal Error Rate (EER), and combined bimodal authentication accuracy using simulated voter datasets.

b. **Scalability and Concurrency Evaluation:** Stress testing from 100 to 10,000 concurrent users using Apache JMeter; comparing database partition efficiency between partitioned and non-partitioned configurations.

c. **Mock Election Simulation:** Executing a simulated election across all nine constituencies, measuring routing correctness, ballot isolation, result integrity, and voter satisfaction.

d. **Voter Satisfaction Assessment:** Evaluating usability through a System Usability Scale (SUS) survey targeting a SUS score of 68 or above.

3.3 System Architecture

The proposed bimodal e-voting system is structured using a client-server architecture designed to ensure robust security, high availability, and efficient processing of electoral data across nine concurrent federal constituency elections. This architecture accommodates mobile application interfaces for voters and an administrative web dashboard for electoral officers. Figure 3.2 presents the full layered architecture of the system.

The client side comprises a React Native mobile application handling voter registration, biometric enrolment, on-device biometric authentication, ballot display, and vote submission. All biometric computations—including fingerprint matching and facial recognition inference—are performed entirely on-device using TensorFlow Lite, ensuring functionality in the low-connectivity environments characteristic of rural Ondo State. The server side consists of a Node.js/Express backend responsible for voter verification, constituency routing logic, encrypted vote storage, and result collation. PostgreSQL with table partitioning forms the data layer, with each constituency and LGA maintaining isolated partitions for voter records, ballots, votes, and collated results, thereby preventing cross-constituency data contamination.

Security is woven throughout every layer. Biometric templates are stored within the device hardware's Trusted Execution Environment (TEE) or Secure Enclave; no raw biometric data is ever transmitted over the network. Votes are encrypted using AES-256 upon submission, and all client-server communications are secured through SSL/TLS protocols. Comprehensive audit trails log all authentication attempts, vote submissions, and administrative actions to ensure auditability and non-repudiation.-

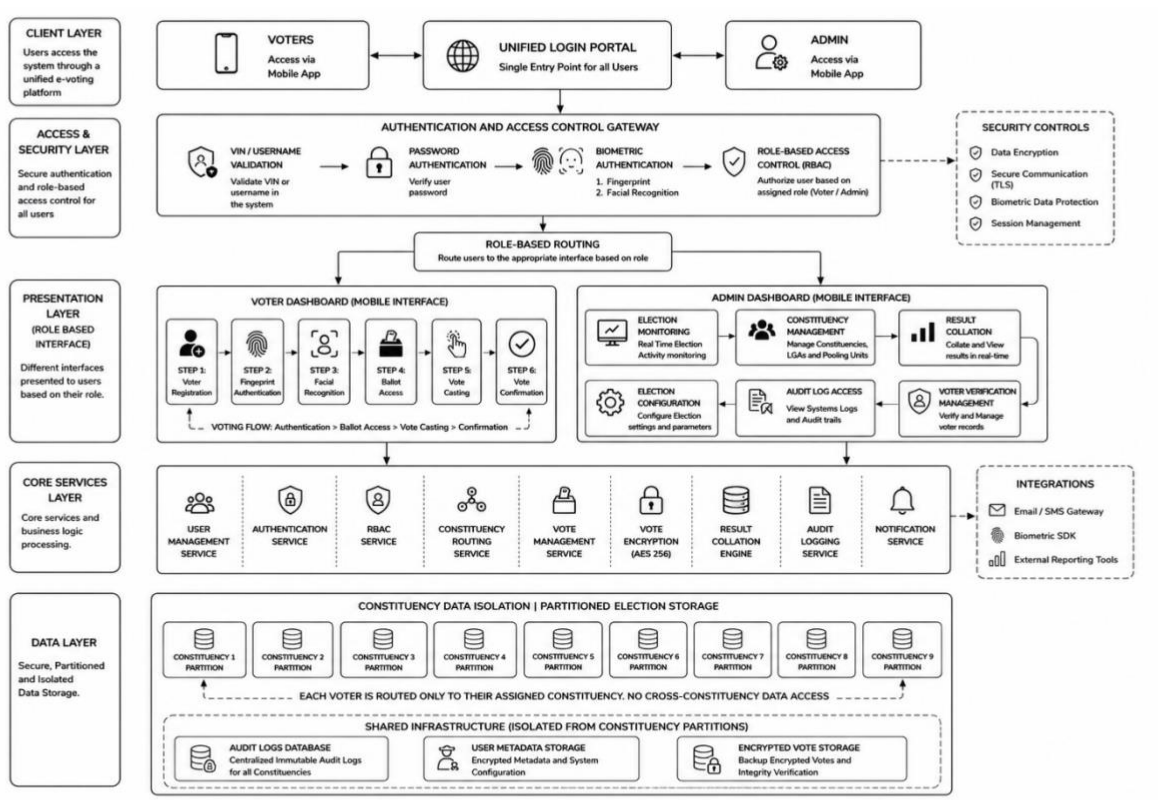

Figure 3.2: System Architecture of the Proposed Bimodal E-Voting System

3.4 Biometric Authentication Design and Implementation

The bimodal e-voting system employs a robust authentication process beginning with knowledge-based verification using a Voter Identification Number (VIN), followed by a strong on-device biometric verification step. The biometric operation follows an inclusive OR logic: successful authentication from either fingerprint or facial recognition grants the voter access to their designated ballot. This design addresses the documented challenge that approximately 24.4 percent of voters with worn fingerprints fail fingerprint-only authentication (Mao et al., 2024), ensuring no eligible voter is disenfranchised by biometric modality failure.

The overall biometric decision rule is expressed formally as:

$$Auth\_result = FP\_auth \ OR \ FR\_auth \qquad \text{(Eq. 3.1)}$$

where FP_auth ∈ {0, 1} is the fingerprint authentication outcome, FR_auth ∈ {0, 1} is the facial recognition outcome, and Auth_result = 1 (access granted) if and only if at least one modality returns a successful match.

3.4.1 Voter Registration

Voters who have not previously registered on the e-voting platform are required to complete the registration process by submitting personal details including full name, email address, phone number, and the Voter Identification Number (VIN) issued automatically and received by the voter via email or SMS. The VIN is validated against the registered voter database to confirm the voter's eligibility and constituency assignment. Upon successful VIN validation, the voter proceeds to biometric enrolment, capturing both fingerprint and facial biometric templates, which are stored securely within the device's TEE and encrypted using AES-256 before any server-side storage. Duplicate detection mechanisms prevent multiple registrations from a single voter. Figure 3.3 illustrates the complete voter registration flowchart.

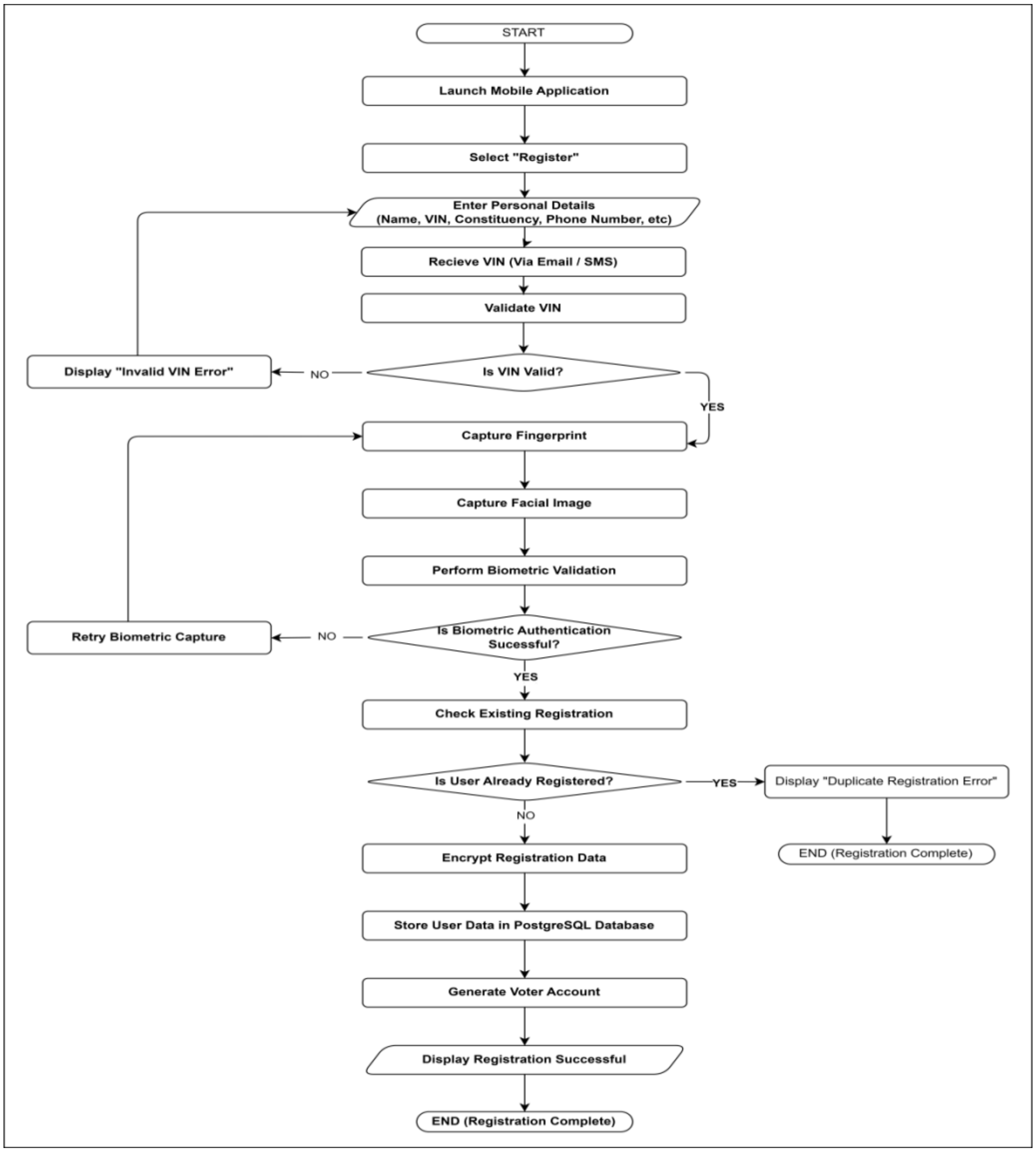

Figure 3.3: Voter Registration Process Flowchart

3.4.2 Fingerprint Authentication Module

The fingerprint authentication module integrates directly with the smartphone's native biometric hardware using the Android BiometricPrompt API, which supports capacitive, optical, and ultrasonic fingerprint sensor types. Templates are stored exclusively within the device's TEE, and only a cryptographic authentication token is returned upon successful verification; raw fingerprint data never leaves the device.

Fingerprint matching is performed using a minutiae-based approach. Each fingerprint template is represented as a set of minutiae points $M = \{m_1, m_2, \ldots, m_n\}$, where each minutia m_i is characterised by its spatial coordinates (x_i, y_i) and ridge orientation θ_i. The matching score S between a live sample L and a stored template T is computed as:

$$S(L, T) = |M\_L \cap M\_T| \,/\, \max(|M\_L|, |M\_T|) \qquad \text{(Eq. 3.2)}$$

where M_L is the set of minutiae points in the live fingerprint sample, M_T is the set of minutiae in the stored template, and |·| denotes set cardinality. A match is accepted when S(L, T) ≥ τ_fp, where τ_fp is the configured fingerprint decision threshold.

To improve resilience for voters with worn fingerprints, the system supports multiple finger enrolment, providing backup authentication options. Authentication logs are maintained for audit purposes while preserving voter anonymity through cryptographic separation of identity verification and vote casting processes.

3.4.3 On-Device Facial Recognition Authentication

The on-device facial recognition module provides an alternative authentication pathway implemented entirely using TensorFlow Lite, comprising a four-stage processing pipeline as illustrated in Figure 3.4 and Figure 3.5.

a. **Face Detection:** A MobileNet-SSD model achieves approximately 30 frames per second on mid-range Android devices, detecting and localising the voter's face in real time.

b. **Landmark Extraction and Alignment:** A 68-point CNN landmark regressor extracts facial keypoints used for pose normalisation and facial alignment, ensuring consistent feature extraction across varying head angles.

c. **Feature Embedding:** A quantised FaceNet model (INT8) maps the aligned facial image to a compact embedding vector. Identity is verified by computing the Euclidean distance between the live embedding and the stored enrolment embedding.

d. **Anti-Spoofing and Liveness Verification:** Eye Aspect Ratio (EAR) temporal analysis and head-rotation challenge-response prompts verify three-dimensional facial motion. A one-class CNN liveness model achieves approximately 99.3 percent spoof detection accuracy against photograph and video-based presentation attacks.

The Euclidean distance between a live embedding vector e_live and the stored template embedding e_stored, used as the facial recognition matching metric, is defined as:

$$d(e\_live,\ e\_stored) = \|e\_live - e\_stored\|_2 = \sqrt{\left(\Sigma_i\ (e\_live_{,i} - e\_stored_{,i})^2 \right)} \quad \text{(Eq. 3.3)}$$

where $e\_live_{,i}$ and $e\_stored_{,i}$ are the i-th components of the live and stored 128-dimensional embedding vectors respectively, $\|\cdot\|_2$ denotes the L_2 (Euclidean) norm, and a match is accepted when $d(e\_live, e\_stored) < \tau\_fr = 0.6$.

The Eye Aspect Ratio (EAR) used for blink-based liveness detection is computed from six facial landmark coordinates per eye as:

$$EAR = (\|p_2 - p_6\| + \|p_3 - p_5\|) \,/\, (2 \cdot \|p_1 - p_4\|) \quad \text{(Eq. 3.4)}$$

where p_1 through p_6 are the six 2-D landmark coordinates of the eye region ordered clockwise. EAR drops sharply during a blink; a value below 0.20 sustained for three consecutive frames is classified as a genuine blink event, confirming liveness.

The bimodal decision logic grants voter access upon successful authentication from either module, ensuring inclusive participation while maintaining biometric security. Figure 3.4 depicts the complete bimodal authentication pipeline, and Figure 3.5 shows the on-device CNN processing stages.

3.5 Voting Process

Following successful biometric authentication and voter accreditation, the system guides the voter through a structured four-stage voting workflow designed to ensure ballot integrity, constituency isolation, and cryptographic security of every submitted vote. The complete voting process is illustrated in Figure 3.4 and is described in detail below.

Stage 1 – Voter Accreditation. The voter launches the mobile application and logs in by entering their Voter Identification Number (VIN) and password. The system proceeds to fingerprint authentication and facial recognition simultaneously, then verifies credentials and biometrics against stored records. If authentication fails, the voter is prompted to retry; upon successful authentication, the session proceeds to the next stage.

Stage 2 – Ballot Access and Verification. Once authenticated, the system performs VIN-based constituency mapping to detect the voter's registered federal constituency. The corresponding constituency-specific election ballot is fetched from the partitioned database, and the candidate list along with ballot instructions is displayed exclusively to the authenticated voter. This ensures strict ballot isolation: no voter can access or view the ballot of another constituency.

Stage 3 – Voting Process. The voter selects their preferred candidate from the displayed list to cast their vote. The system presents a review screen showing the selected choice, after which the voter is asked to confirm. If the voter chooses not to confirm, they are returned to the candidate selection screen to modify their choice. Upon confirmation, the vote proceeds to the submission stage.

Stage 4 – Submission and Confirmation. The confirmed vote undergoes server-side validation against business rules and voter eligibility constraints. The valid vote record is then encrypted using AES-256 in CBC mode and transmitted to the backend over a secure SSL/TLS channel. The encrypted vote is stored in the constituency-specific database partition. A unique Vote ID is generated and a confirmation receipt is displayed to the voter, concluding the voting session.

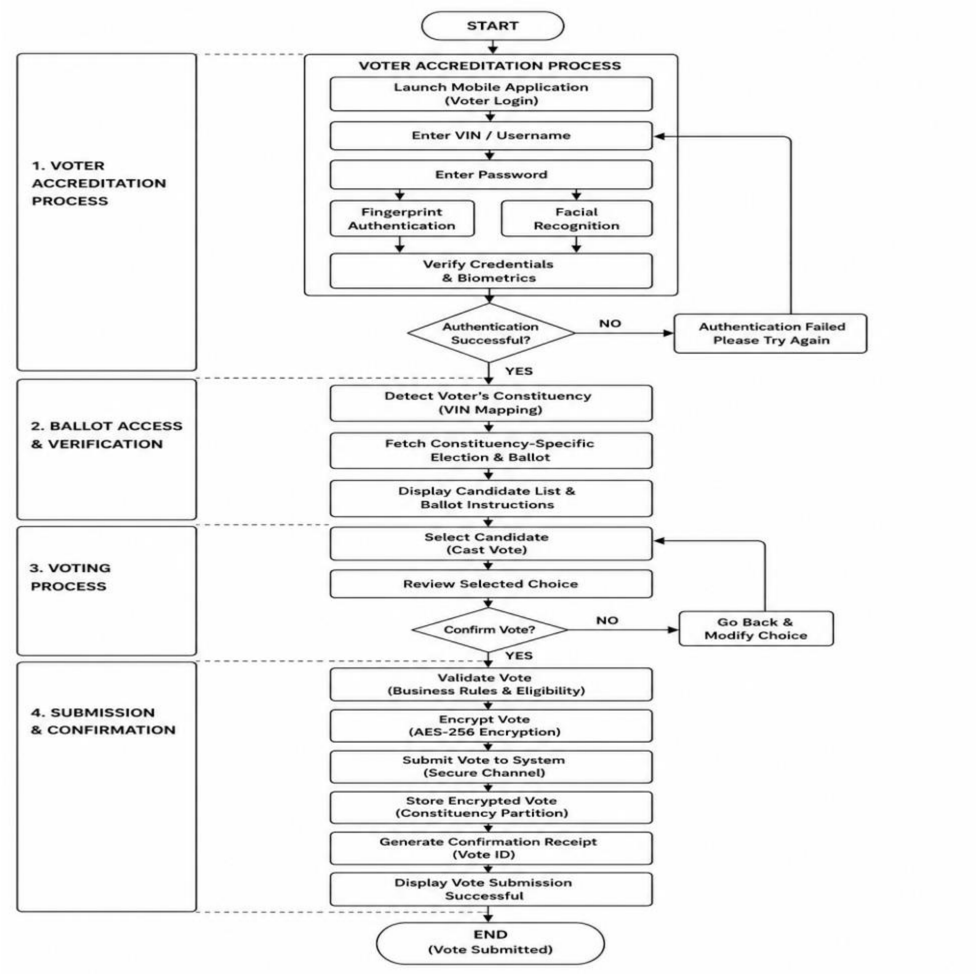

Figure 3.4: Voting Process Flowchart

3.5 Multi-Constituency and Concurrent Election Architecture

A defining architectural requirement of this system is the ability to manage nine simultaneous House of Representatives elections with complete data isolation across all nine Federal Constituencies and eighteen Local Government Areas of Ondo State. This is achieved through a constituency-aware voter routing mechanism and a partitioned database design, as illustrated in Figure 3.6.

Upon successful biometric authentication, the system queries the voter's VIN to determine their registered constituency and routes them exclusively to the corresponding ballot. The routing function R(v) is defined formally as:

$$R(v) = C\_k \quad \text{such that} \quad VIN(v) \in \Omega\_k, \quad k \in \{1, 2, \ldots, 9\} \qquad \text{(Eq. 3.5)}$$

where v is the authenticated voter, VIN(v) is the voter's identification number, Ω_k is the set of VINs registered to constituency C_k, and R(v) returns the unique constituency partition assigned to voter v. If VIN(v) ∉ ⋃Ω_k the request is rejected.

The application enforces independent ballot access at both the API and database layers: voters are presented only with the candidate list associated with their constituency, and any attempt to access another constituency's ballot is rejected at multiple validation checkpoints. Result collation is similarly partitioned, with each constituency maintaining independent vote totals, collation records, and transmission streams.

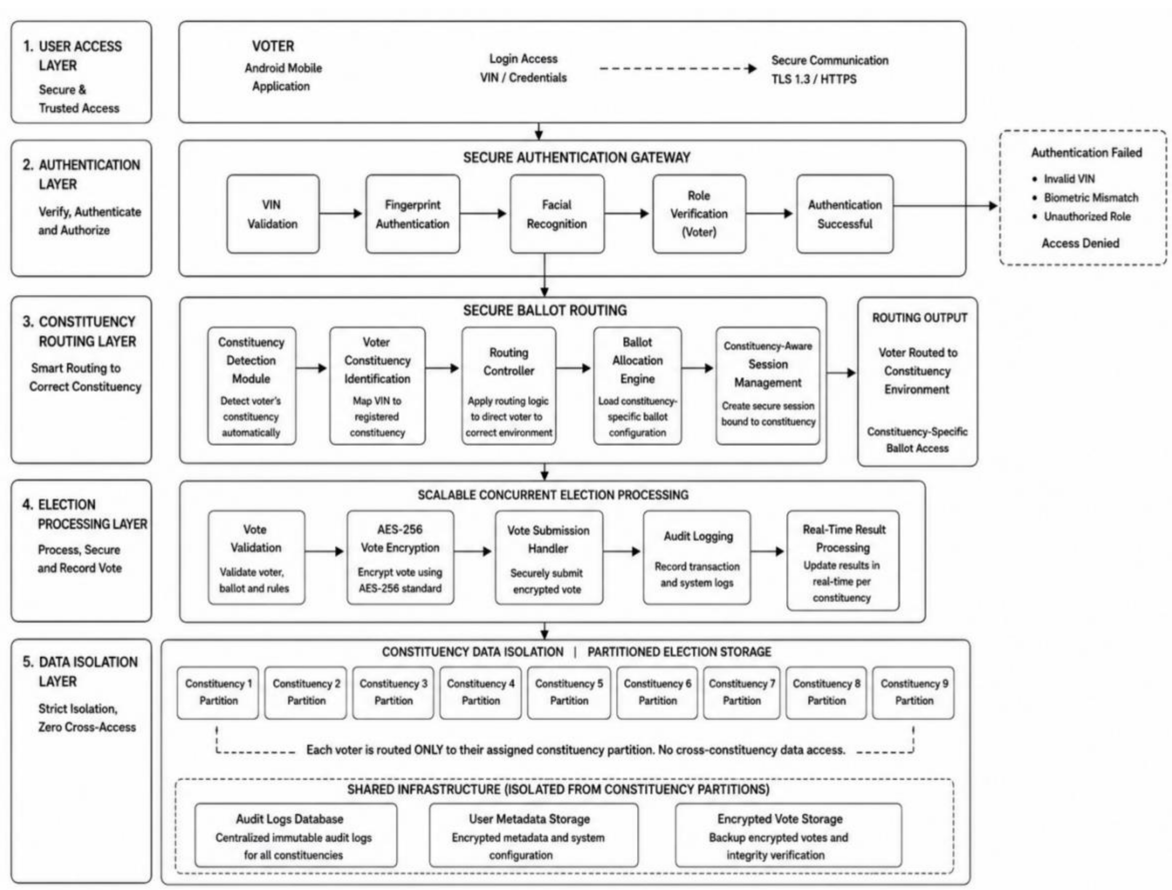

Figure 3.5: Constituency-Aware Voter Routing and Data Isolation Architecture

3.6 Other System Implementation Details

Beyond biometric authentication and multi-constituency management, the system's implementation encompasses robust modules for data management, vote processing, and overall security.

The backend logic is developed using Node.js with Express, chosen for its asynchronous, event-driven processing capabilities suited to high-concurrency election scenarios. PostgreSQL with table partitioning serves as the DBMS, storing all voter registration data, encrypted votes, and audit logs. The database partitioning scheme isolates records per constituency, ensuring that queries for Constituency k access only the partition Ω_k, reducing contention and improving query latency during peak voting.

Vote encryption is performed using AES-256 in Cipher Block Chaining (CBC) mode. Each submitted vote record V is encrypted as:

$$C = AES\text{-}256\_CBC(V,\ K\_session,\ IV) \qquad \text{(Eq. 3.6)}$$

where C is the resulting ciphertext stored in the database, V is the plaintext vote record, K_session is a per-session 256-bit symmetric encryption key derived via HKDF, and IV is a randomly generated 128-bit initialisation vector unique to each vote. Both K_session and IV are discarded after the session, ensuring forward secrecy.

All client-server communications are encrypted via SSL/TLS. Comprehensive audit trail mechanisms log all authentication attempts, vote casting events, and administrative data-access operations, ensuring non-repudiation. The architecture emphasises an offline-first design: on-device TensorFlow Lite inference reduces server load during peak voting periods and ensures authentication reliability even in areas with limited network connectivity.

3.7 Administrative Operational Workflow

The e-voting system is not solely a voter-facing application; it encompasses a complete election management platform operated by authorised electoral officers (Admins). The administrative workflow governs the full lifecycle of an election, from configuration and candidate upload through real-time monitoring, voter oversight, result collation, and audit control. Figure 3.6 illustrates the complete administrative operational workflow.

Access to the administrative dashboard is protected by a multi-factor authentication mechanism requiring the electoral officer's Admin ID, password, and biometric verification, followed by role-based access control (RBAC) enforcement. Only authenticated administrators whose roles permit the requested operation are granted access; all other attempts terminate with an "Access Denied" response and are logged in the audit trail.

Upon successful authentication, the administrator interacts with six functional modules. The Election Management Module enables configuration of election settings, upload of candidate information, assignment of elections to their respective constituencies, and activation of election sessions. The Voter Management Module supports verification of registered voters, monitoring of registration activities, detection of duplicate registrations, and management of voter records. The Constituency Management Module handles configuration of constituency partitions, management of LGA allocations, and monitoring of ballot routing integrity to ensure that no voter is routed to an incorrect constituency ballot.

The Election Monitoring Module provides real-time visibility into voting activities across all nine federal constituencies simultaneously, monitors for concurrent election anomalies, and detects suspicious activities such as abnormally high vote rates or repeated failed authentication attempts. The Result Collation Module fetches constituency-level vote totals, verifies vote integrity against the encrypted records in the partitioned database, collates election results across all constituencies, and generates official election reports. Finally, the Audit and Security Module exposes administrative audit logs, supports review of all administrative activities for accountability, and enables backup of encrypted records to ensure data durability and compliance with electoral record-keeping requirements.

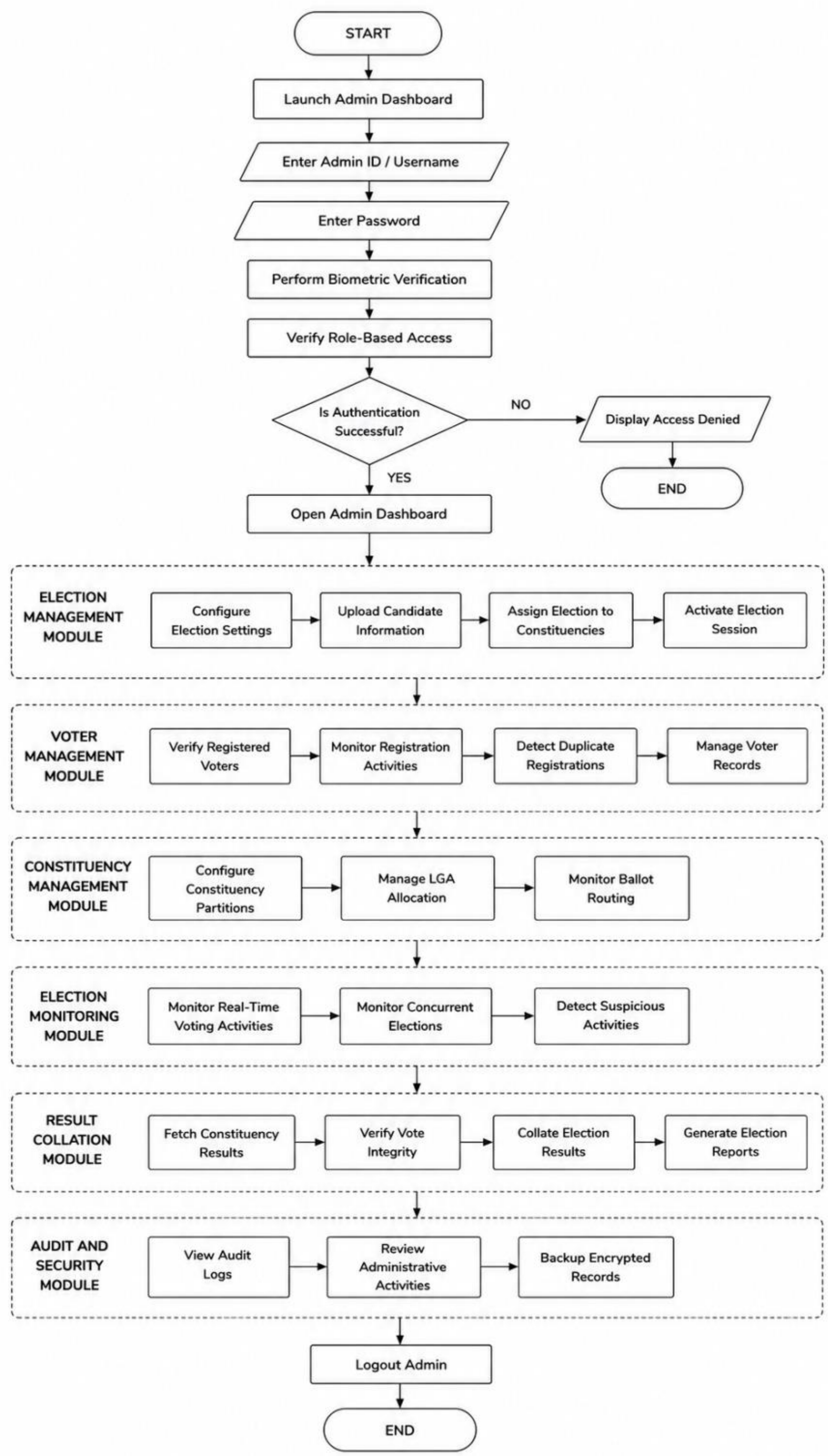

Figure 3.6: Administrative Operational Workflow Flowchart

3.8 Performance Evaluation Metrics and Setup

The performance evaluation phase will rigorously test the implemented bimodal e-voting system against specific metrics to ensure its reliability, efficiency, accuracy, and operational scalability for real-world electoral deployment.

3.8.1 Performance Metrics

The following metrics will be measured during the evaluation phase:

- Average Response Time: Average time taken for key operations including login, biometric verification, and vote casting.
- Throughput: Number of successful authentications or votes processed per unit time, assessing capacity for large-scale elections.
- Biometric Verification Accuracy: FAR (target < 0.01%), FRR for facial recognition (target < 1.5%), FRR for fingerprint (target < 0.8%), EER, and combined bimodal accuracy (target ≥ 99.7%).
- Routing Correctness: Verified via simulation of 10,000 voters across nine constituencies, targeting zero ballot leakage incidents.
- Database Partition Efficiency: Query response times for partitioned vs. non-partitioned configurations, targeting response below 500 ms under peak load.
- System Usability Scale (SUS): Targeting a SUS score ≥ 68 across diverse user demographics.

The Equal Error Rate (EER), which represents the single-value operating point where the False Acceptance Rate equals the False Rejection Rate, is defined as:

$$EER = \tau^* \quad \text{such that} \quad FAR(\tau^*) = FRR(\tau^*) \qquad \text{(Eq. 3.7)}$$

where τ^* is the optimal decision threshold at which FAR and FRR are equal. $FAR(\tau) = FP / (FP + TN)$ is the proportion of unauthorised users incorrectly accepted, and $FRR(\tau) = FN / (FN + TP)$ is the proportion of legitimate voters incorrectly rejected. A lower EER indicates superior biometric system performance.

The System Usability Scale (SUS) score used to evaluate user experience is computed from a 10-item Likert questionnaire as:

$$SUS = 2.5 \times [\ \Sigma(\text{odd item scores} - 1) + \Sigma(5 - \text{even item scores})\ ] \quad \text{(Eq. 3.8)}$$

where odd item scores are responses to positively-worded items (Q1, Q3, Q5, Q7, Q9) and even item scores are responses to negatively-worded items (Q2, Q4, Q6, Q8, Q10), each rated on a 1–5 Likert scale. The resulting SUS score ranges from 0 to 100; a score ≥ 68 is considered above average usability (Brooke, 1996).

3.8.2 Testing Setup

The system will be tested as a prototype in a controlled environment. Simulated voter datasets covering all nine Federal Constituencies and eighteen LGAs will be generated, incorporating controlled variations in biometric quality to assess system robustness across diverse demographics including elderly and agricultural workers with worn fingerprints. Load and stress testing will be conducted using Apache JMeter, simulating concurrent authentication requests from 100 to 10,000 users to determine maximum capacity and stability under peak election conditions. Usability testing will involve a mock election simulation with participants drawn from diverse age groups and educational backgrounds, collecting SUS scores and qualitative feedback. Security assessment will include vulnerability analysis, cross-constituency access injection testing, and injection of invalid constituency votes to verify multi-layer validation controls.

3.9 Ethical and Legal Considerations

Throughout the project, significant attention will be paid to the ethical implications and legal compliance of the system, particularly concerning biometric data handling and electoral laws in Nigeria. The system's design adheres to the Nigeria Data Protection Regulation (NDPR), ensuring transparency, securing informed consent for biometric data collection, and implementing mechanisms to prevent misuse of sensitive voter information. No raw biometric data is stored on external servers; all biometric processing occurs on-device within hardware-isolated security enclaves, ensuring that template reconstruction remains computationally infeasible even in the event of a database breach. The system's design integrates features aligned with the principles of voter privacy, ballot anonymity, and the overall integrity of the democratic process, consistent with the requirements of the Nigerian Electoral Act (2022).